Homework 2, Business 33320, University of Chicago Booth School of Business
Due Week 4 (April 14, 2026)

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1. Consider the 3-node electricity network given in class in Week 2. If you use Excel, you may use the 3nodelp.xlsx file on Canvas. For the questions below, assume that the generator at Node 1 is a wind generator with zero marginal cost of production and that the capacity of the generator at Node 2 is 5 MW's, and that the load at Node 3 is also 5 MW's.

a) Suppose the available wind generation at Node 1 is 5 MW's. What is the amount of wind generation and Node 2 generation in this case? What would be the LMP's at each node? How much of the zero cost wind power is curtailed (i.e., notavailable) due to the transmission capacity?

b) Suppose now that the wind generation capacity at Node 1 is random and normallydistributed with a mean of 5 MW's and a standard deviation of 2 MW's. Generate30 scena rios of wind generation capacity and find the generation levels at Nodes 1 and 2, total cost for one hour, LMP's, and the amount of wind curtailment for each scenario. You may not be able to meet the load in every scenario. In the event that wind capacity is insufficient to meet the load, value the lost load at a cost of $1000/MW-hour. To do this, you can add an additional decision variable in the linear program for lost load on the left-hand side of the Node 3 flow constraint.

c) Using the same scenarios as you generated for (b), increase the capacity of Line 1 transmission to 1.5 MW's. With this expanded transmission capacity, repeat your calculations of the generation at Nodes 1 and 2, total cost for one hour, LMP's, and the amount of wind curtailment for each scenario.

d) Using your observations above, assuming these conditions occur during 3000 peak hours in a year, what would be your estimate of the annual value of increasing the capacity of Line 1 from 1 MW to 1.5 MW and by what fraction does this transmission increase the overall efficiency of a wind generator investment (i.e., increase in the fraction of the available wind capacity that is able to be used)?

2. Download the data for February 2025 for the MISO LMP's from the MISO site. The files are given ashttps://docs.misoenergy.org/marketreports/YYYYMMD YYYYMMDD stands for the year, month, and day.

Programatically downloading the MISO real-time LMP final files follow the URL pattern, we have: `https://docs.misoenergy.org/mar`

```
#once run, no need to re-download files
dir.create(data_dir, showWarnings = FALSE)

feb_dates <- seq(as.Date("2025-02-01"), as.Date("2025-02-28"), by = "day")
```

```r
download_lmp <- function(date) {
  date_str  <- format(date, "%Y%m%d")
  url       <- paste0("https://docs.misoenergy.org/marketreports/",
                      date_str, "_rt_lmp_final.csv")
  dest_file <- file.path(data_dir, paste0(date_str, "_rt_lmp_final.csv"))
  if (!file.exists(dest_file)) {
    tryCatch(
      download.file(url, dest_file, quiet = TRUE, mode = "wb"),
      error   = function(e) message("Download failed: ", date_str),
      warning = function(w) NULL
    )
  }
  dest_file
}

invisible(lapply(feb_dates, download_lmp))
```

Inspecting file structure:

```r
files <- list.files(data_dir, pattern = "_rt_lmp_final\\.csv$", full.names = TRUE)
cat("Files present:", length(files), "\n")
```

```
## Files present: 28
```

```r
# The MISO file is wide-format: rows = nodes, columns = HE 1 .. HE 24
# Header is on row 5 (skip 4 rows).
raw_head <- read_csv(files[1], skip = 4, n_max = 3, show_col_types = FALSE)
print(names(raw_head))
```

```
##  [1] "Node"  "Type"  "Value" "HE 1"  "HE 2"  "HE 3"  "HE 4"  "HE 5"  "HE 6"
## [10] "HE 7"  "HE 8"  "HE 9"  "HE 10" "HE 11" "HE 12" "HE 13" "HE 14" "HE 15"
## [19] "HE 16" "HE 17" "HE 18" "HE 19" "HE 20" "HE 21" "HE 22" "HE 23" "HE 24"
```

```r
print(raw_head[, 1:6])
```

```
## # A tibble: 3 x 6
##   Node      Type      Value `HE 1` `HE 2` `HE 3`
##   <chr>     <chr>     <chr>  <dbl>  <dbl>  <dbl>
## 1 AECI      Interface LMP     24.5   24.2   19.4
## 2 AECI.ALTW Loadzone  LMP     24.0   31.9   29.2
## 3 AECI.AMMO Loadzone  LMP     24.2   24.1   30.1
```

Parsing each file to long format:

```r
read_lmp_file <- function(path) { # read one day's file and return node x hour x LMP rows
  date_str <- str_extract(basename(path), "^[0-9]{8}")
  date_val <- as.Date(date_str, "%Y%m%d")

  # Skip 4-line preamble; header row is row 5
  df <- read_csv(path, skip = 4, show_col_types = FALSE)

  # Keep only LMP rows (Value column == "LMP")
  # Columns: Node, Type, Value, HE 1, HE 2, ..., HE 24
  df <- df %>% filter(Value == "LMP")

  # Identify the HE columns
```

```r
  he_cols <- names(df)[str_detect(names(df), "^HE\\s*[0-9]+$")]

  # Pivot to long: one row per (node, hour)
  df_long <- df %>%
    select(Node, all_of(he_cols)) %>%
    pivot_longer(
      cols      = all_of(he_cols),
      names_to  = "hour_label",
      values_to = "lmp"
    ) %>%
    mutate(
      hour     = as.integer(str_extract(hour_label, "[0-9]+")),
      lmp      = as.numeric(lmp),
      date     = date_val,
      day_type = if_else(wday(date_val, week_start = 1) <= 5,
                         "Weekday", "Weekend")
    ) %>%
    select(date, day_type, hour, lmp) %>%
    filter(!is.na(lmp))

  df_long
}

all_lmp <- map_dfr(files, read_lmp_file)

cat("Total node-hour observations:", nrow(all_lmp), "\n")
```

```
## Total node-hour observations: 1655808
```

```r
cat("Distinct dates:", n_distinct(all_lmp$date), "\n")
```

```
## Distinct dates: 28
```

```r
cat("Day-type breakdown:\n")
```

```
## Day-type breakdown:
```

```r
print(table(format(all_lmp$date[!duplicated(all_lmp$date)],
                   "%Y-%m-%d"),
            all_lmp$day_type[!duplicated(all_lmp$date)]))
```

```
##
##              Weekday Weekend
##   2025-02-01       0       1
##   2025-02-02       0       1
##   2025-02-03       1       0
##   2025-02-04       1       0
##   2025-02-05       1       0
##   2025-02-06       1       0
##   2025-02-07       1       0
##   2025-02-08       0       1
##   2025-02-09       0       1
##   2025-02-10       1       0
##   2025-02-11       1       0
##   2025-02-12       1       0
##   2025-02-13       1       0
##   2025-02-14       1       0
```

```
##   2025-02-15       0       1
##   2025-02-16       0       1
##   2025-02-17       1       0
##   2025-02-18       1       0
##   2025-02-19       1       0
##   2025-02-20       1       0
##   2025-02-21       1       0
##   2025-02-22       0       1
##   2025-02-23       0       1
##   2025-02-24       1       0
##   2025-02-25       1       0
##   2025-02-26       1       0
##   2025-02-27       1       0
##   2025-02-28       1       0
```

Summary stats: Per-day, per-hour across all nodes

For each (date, hour) pair we compute the mean, maximum, minimum, and standard deviation of LMPs across all MISO nodes.

```r
daily_hourly_stats <- all_lmp %>%
  group_by(date, day_type, hour) %>%
  summarise(
    mean_lmp = mean(lmp, na.rm = TRUE),
    max_lmp  = max(lmp,  na.rm = TRUE),
    min_lmp  = min(lmp,  na.rm = TRUE),
    sd_lmp   = sd(lmp,   na.rm = TRUE),
    .groups  = "drop"
  )

glimpse(daily_hourly_stats)
```

```
## Rows: 672
## Columns: 7
## $ date     <date> 2025-02-01, 2025-02-01, 2025-02-01, 2025-02-01, 2025-02-01, ~
## $ day_type <chr> "Weekend", "Weekend", "Weekend", "Weekend", "Weekend", "Weeke~
## $ hour     <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18~
## $ mean_lmp <dbl> 21.23404, 18.61579, 19.23328, 19.58401, 19.44119, 20.67605, 2~
## $ max_lmp  <dbl> 29.47, 38.35, 104.80, 141.27, 157.88, 101.36, 78.92, 109.08, ~
## $ min_lmp  <dbl> 5.83, -22.37, -32.66, -39.70, -42.66, -74.28, -97.27, -82.33,~
## $ sd_lmp   <dbl> 3.488940, 6.653871, 11.501714, 16.422823, 16.868850, 12.85675~
```

Summary stats: Group averages by day type and hour

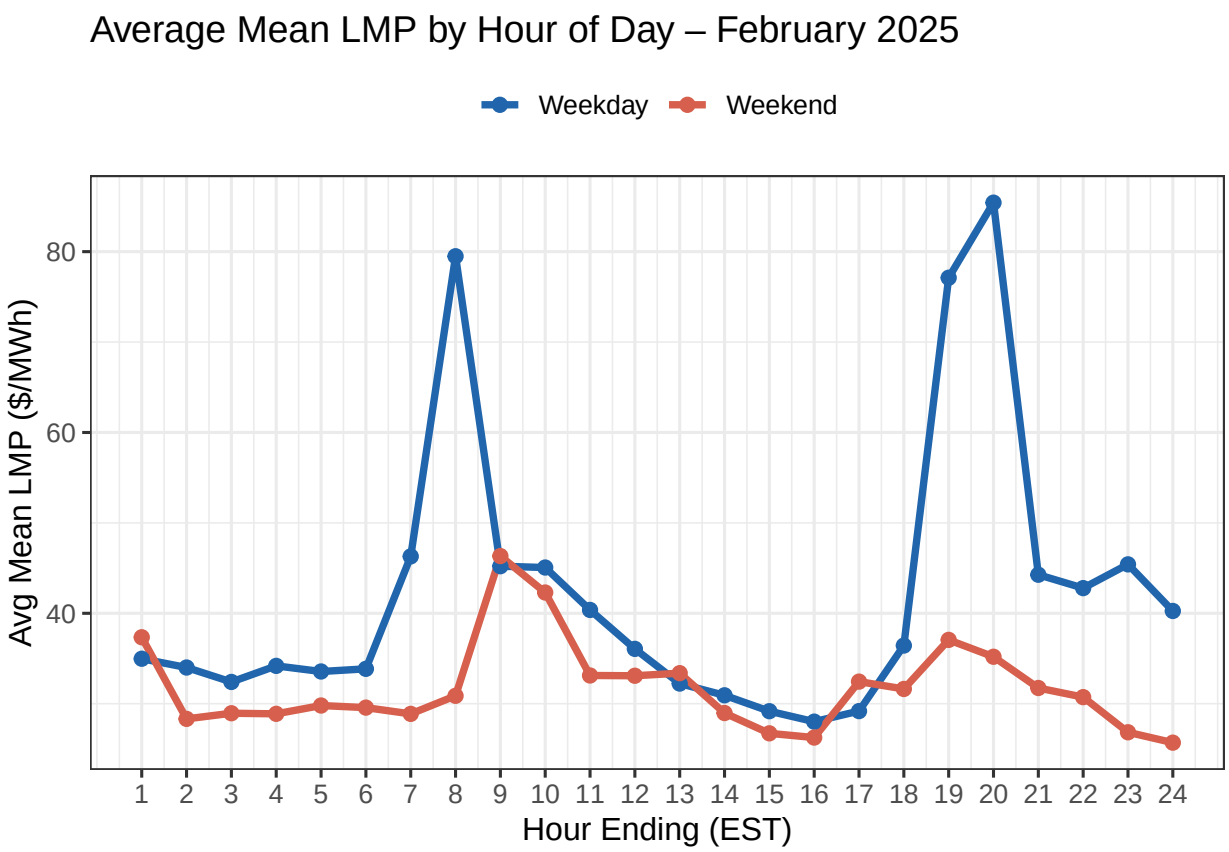
Within each group (Weekday / Weekend) we average each statistic over all days.

```r
group_hourly_avgs <- daily_hourly_stats %>%
  group_by(day_type, hour) %>%
  summarise(
    avg_mean_lmp = mean(mean_lmp, na.rm = TRUE),
    avg_max_lmp  = mean(max_lmp,  na.rm = TRUE),
    avg_min_lmp  = mean(min_lmp,  na.rm = TRUE),
    avg_sd_lmp   = mean(sd_lmp,   na.rm = TRUE),
    .groups = "drop"
  )
```

Plots

Average Mean LMP by Hour of Day

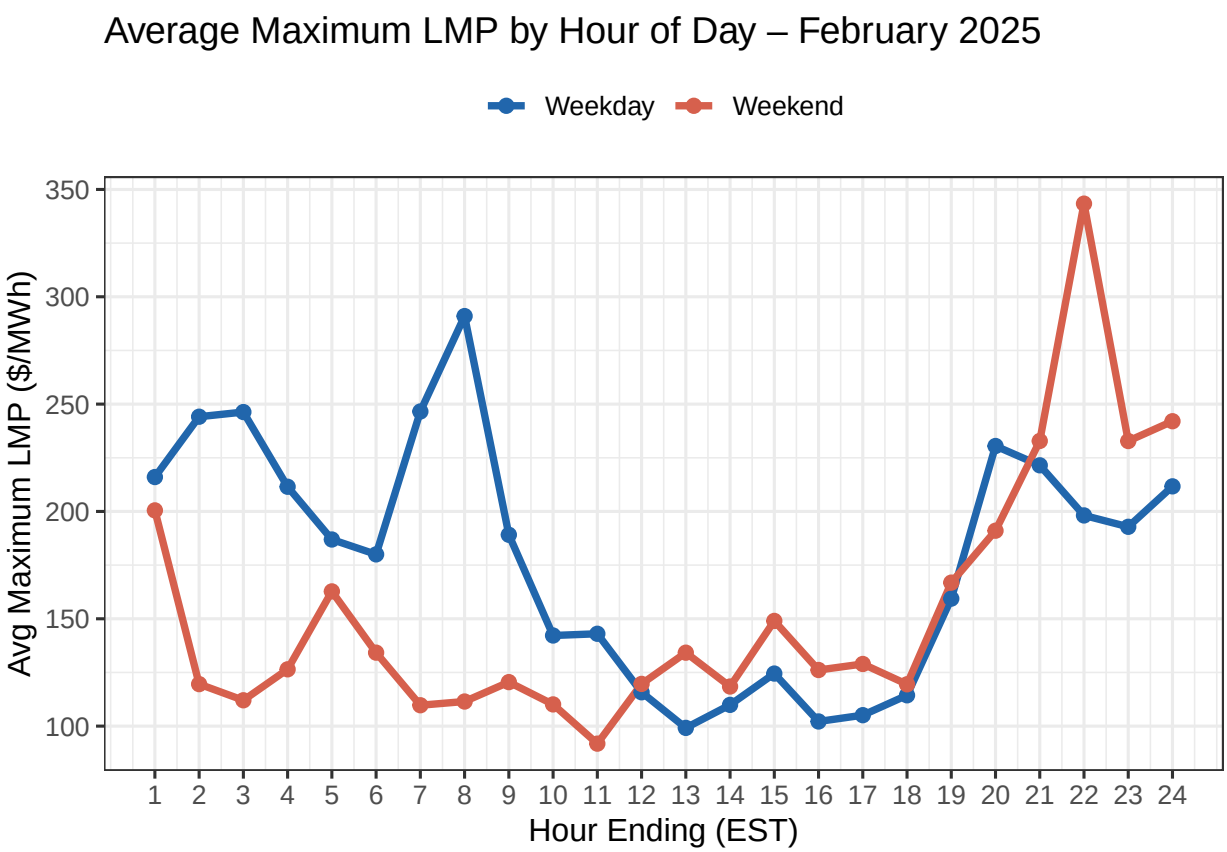
```
ggplot(group_hourly_avgs,
       aes(x = hour, y = avg_mean_lmp, color = day_type, group = day_type)) +
  geom_line(linewidth = 1.3) +
  geom_point(size = 2.2) +
  scale_x_continuous(breaks = 1:24) +
  scale_color_manual(values = c("Weekday" = "#2166ac", "Weekend" = "#d6604d")) +
  labs(
    title = "Average Mean LMP by Hour of Day - February 2025",
    x = "Hour Ending (EST)", y = "Avg Mean LMP ($/MWh)", color = NULL
  ) +
  theme_bw(base_size = 12) +
  theme(legend.position = "top")
```



Average Maximum LMP by Hour of Day

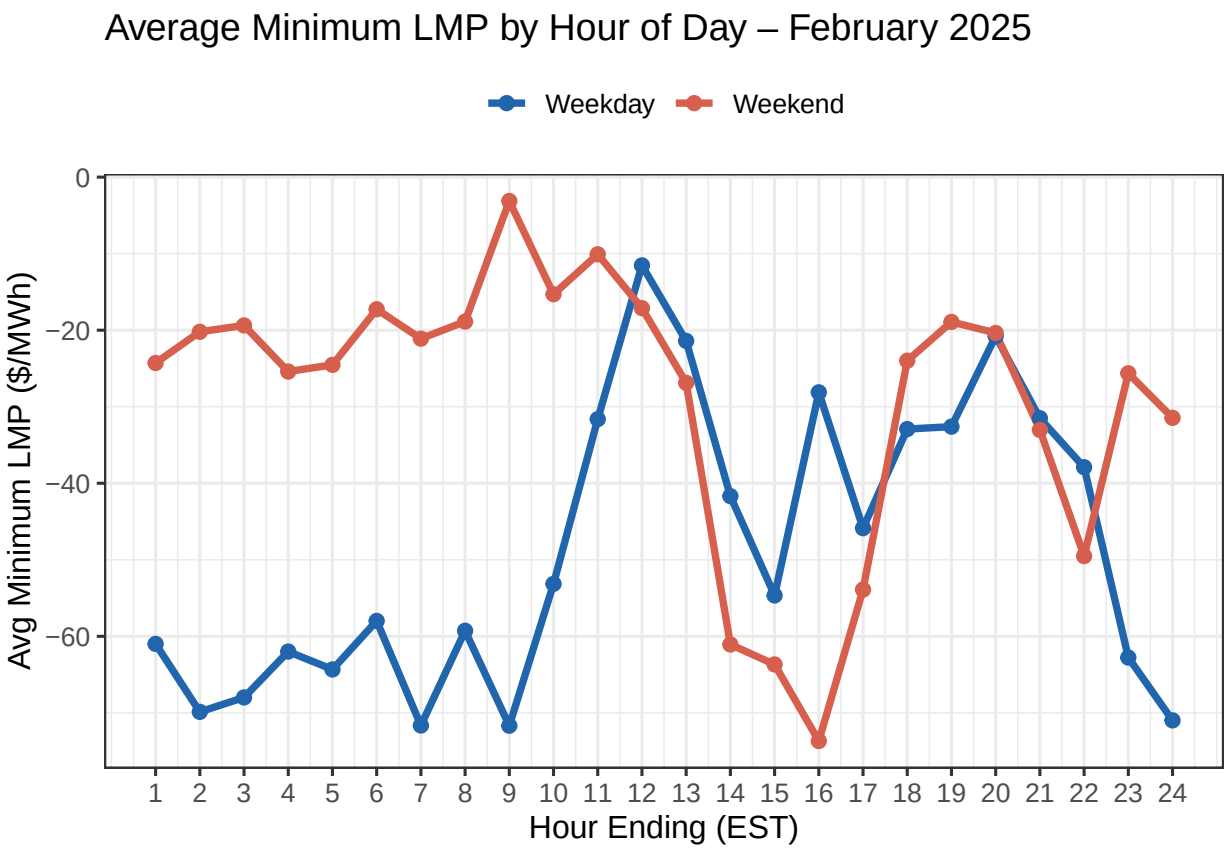
```
ggplot(group_hourly_avgs,
       aes(x = hour, y = avg_max_lmp, color = day_type, group = day_type)) +
  geom_line(linewidth = 1.3) +
  geom_point(size = 2.2) +
  scale_x_continuous(breaks = 1:24) +
  scale_color_manual(values = c("Weekday" = "#2166ac", "Weekend" = "#d6604d")) +
  labs(
    title = "Average Maximum LMP by Hour of Day - February 2025",
    x = "Hour Ending (EST)", y = "Avg Maximum LMP ($/MWh)", color = NULL
  ) +
```

```
  theme_bw(base_size = 12) +
  theme(legend.position = "top")
```



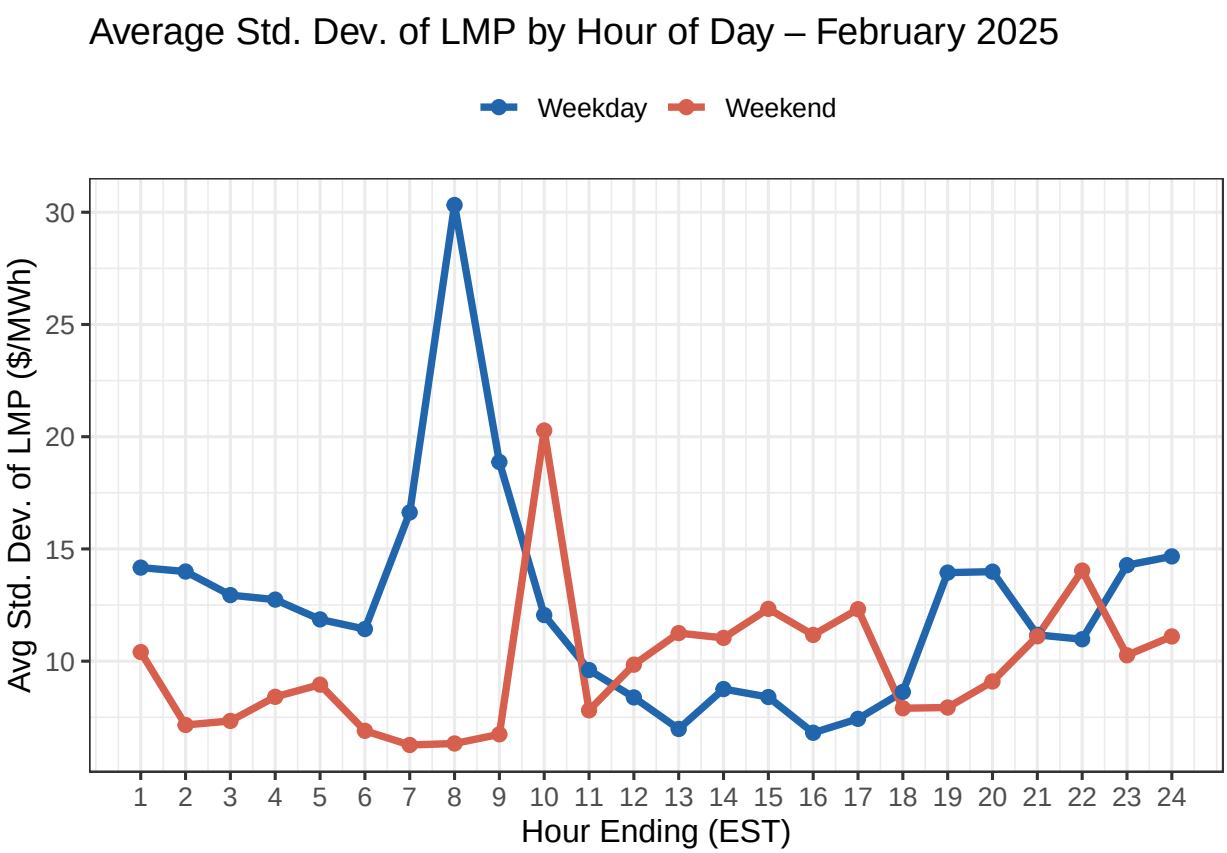
Average Minimum LMP by Hour of Day

```
ggplot(group_hourly_avgs,
       aes(x = hour, y = avg_min_lmp, color = day_type, group = day_type)) +
  geom_line(linewidth = 1.3) +
  geom_point(size = 2.2) +
  scale_x_continuous(breaks = 1:24) +
  scale_color_manual(values = c("Weekday" = "#2166ac", "Weekend" = "#d6604d")) +
  labs(
    title = "Average Minimum LMP by Hour of Day - February 2025",
    x = "Hour Ending (EST)", y = "Avg Minimum LMP ($/MWh)", color = NULL
  ) +
  theme_bw(base_size = 12) +
  theme(legend.position = "top")
```



Average Standard Deviation of LMP by Hour of Day

```
ggplot(group_hourly_avgs,
       aes(x = hour, y = avg_sd_lmp, color = day_type, group = day_type)) +
  geom_line(linewidth = 1.3) +
  geom_point(size = 2.2) +
  scale_x_continuous(breaks = 1:24) +
  scale_color_manual(values = c("Weekday" = "#2166ac", "Weekend" = "#d6604d")) +
  labs(
    title = "Average Std. Dev. of LMP by Hour of Day - February 2025",
    x = "Hour Ending (EST)", y = "Avg Std. Dev. of LMP ($/MWh)", color = NULL
  ) +
  theme_bw(base_size = 12) +
  theme(legend.position = "top")
```



Summary Table

```
knitr::kable(
  group_hourly_avgs %>%
    arrange(day_type, hour) %>%
    mutate(across(where(is.numeric), \(x) round(x, 2))),
  col.names = c("Day Type", "Hour", "Avg Mean LMP",
                "Avg Max LMP", "Avg Min LMP", "Avg SD LMP"),
  caption = "Averaged hourly LMP statistics by day type, MISO February 2025"
)
```

Table 1: Averaged hourly LMP statistics by day type, MISO February 2025

Day Type	Hour	Avg Mean LMP	Avg Max LMP	Avg Min LMP	Avg SD LMP
Weekday	1	34.97	216.02	-60.99	14.17
Weekday	2	34.00	244.13	-69.87	13.99
Weekday	3	32.41	246.30	-67.98	12.94
Weekday	4	34.18	211.50	-61.99	12.74
Weekday	5	33.56	186.93	-64.33	11.86
Weekday	6	33.86	180.00	-57.99	11.43
Weekday	7	46.30	246.57	-71.65	16.63
Weekday	8	79.49	291.01	-59.27	30.32
Weekday	9	45.21	189.13	-71.68	18.87
Weekday	10	45.06	142.21	-53.15	12.05
Weekday	11	40.37	143.01	-31.62	9.61
Weekday	12	36.06	115.76	-11.53	8.39
Weekday	13	32.23	99.17	-21.40	6.98

Day Type	Hour	Avg Mean LMP	Avg Max LMP	Avg Min LMP	Avg SD LMP
Weekday	14	30.93	109.90	-41.69	8.76
Weekday	15	29.17	124.48	-54.66	8.40
Weekday	16	28.01	102.16	-28.12	6.81
Weekday	17	29.18	105.09	-45.87	7.43
Weekday	18	36.43	114.36	-32.92	8.63
Weekday	19	77.11	159.39	-32.60	13.94
Weekday	20	85.42	230.49	-20.78	13.99
Weekday	21	44.26	221.50	-31.51	11.17
Weekday	22	42.78	198.14	-37.90	10.98
Weekday	23	45.41	192.84	-62.77	14.28
Weekday	24	40.26	211.69	-70.98	14.67
Weekend	1	37.35	200.49	-24.28	10.41
Weekend	2	28.32	119.57	-20.21	7.16
Weekend	3	28.94	112.04	-19.37	7.34
Weekend	4	28.88	126.46	-25.41	8.41
Weekend	5	29.80	162.76	-24.53	8.95
Weekend	6	29.57	134.22	-17.26	6.90
Weekend	7	28.87	109.72	-21.10	6.27
Weekend	8	30.87	111.46	-18.87	6.33
Weekend	9	46.33	120.49	-3.12	6.74
Weekend	10	42.30	110.11	-15.30	20.27
Weekend	11	33.12	91.83	-10.09	7.81
Weekend	12	33.09	119.65	-17.12	9.84
Weekend	13	33.38	134.21	-26.88	11.25
Weekend	14	28.97	118.44	-61.07	11.04
Weekend	15	26.72	149.00	-63.68	12.33
Weekend	16	26.25	126.14	-73.68	11.17
Weekend	17	32.44	128.94	-53.89	12.32
Weekend	18	31.63	119.48	-23.99	7.90
Weekend	19	37.06	166.84	-18.92	7.94
Weekend	20	35.19	191.03	-20.36	9.10
Weekend	21	31.74	232.83	-33.03	11.11
Weekend	22	30.73	343.38	-49.51	14.04
Weekend	23	26.84	232.79	-25.63	10.26
Weekend	24	25.69	242.03	-31.45	11.10

Part b)

Here are my observations on these metrics as a function of hour-of-day:

- **Mean LMP (weekdays).** Overnight (HE 1–6) weekday mean LMPs average roughly 33.5 $/MWh, reflecting cheap baseload (coal, nuclear) supplying modest night-time demand. Prices rise sharply through the morning ramp as commercial and industrial loads come online, reaching a first peak near HE 20 at about 85.4 $/MWh. This is consistent with the "Weekly Load Variation" slide, which shows peak demand mid-week during business hours. A second, often comparable evening peak appears around HE 18–20 as residential demand surges after working hours before falling through the night.
- **Mean LMP (weekends).** Weekend mean LMPs are uniformly lower and their daily profile is flatter. The morning ramp is delayed by 1–2 hours (later commercial activity) and the midday dip is shallower. The absence of large industrial loads during weekends suppresses prices relative to weekdays for the same calendar month and temperature regime.
- **Maximum LMP.** The average maximum LMP—representing the highest-priced node in the MISO footprint each hour—is considerably higher than the mean, especially during weekday peak hours. These spikes reflect (i) the merit-order effect as expensive peakers are dispatched when load crowds lower-cost units, and (ii) transmission congestion that isolates high-demand nodes, as illustrated in the slide showing $105/MWh vs. -$30/MWh nodes only 20 miles apart. Weekend maximum LMPs show smaller spikes and track closer to the mean.
- **Minimum LMP.** Average minimum LMPs reach their lowest values (-71.7 $/MWh on weekdays and -73.7 $/MWh on weekends) during overnight hours. Negative values arise when surplus generation—wind in particular, which has zero

marginal cost and cannot be cheaply curtailed— floods the grid at times of low demand. This mirrors the overgeneration risk discussed in the Duck Curve and Merit-Order slides: when wind displaces dispatchable units at the bottom of the merit order, nodes with export constraints can see prices go negative.

- **Standard Deviation of LMPs.** The cross-sectional standard deviation quantifies how much prices diverge *between nodes* during each hour—a direct proxy for the magnitude of transmission congestion. On weekdays the standard deviation peaks around HE 8 at 30.3 $/MWh, while weekends peak at only 20.3 $/MWh. This confirms that transmission lines are most severely congested during weekday peak hours, creating the widest node-to-node price spreads.

Implications for Financial Transmission Rights (FTRs)

An FTR between nodes A and B pays the holder $(LMP_B - LMP_A) \times MWh$ nominated. The payoff is positive when B is the expensive (load-side) node and A is the cheap (generation-side) node; it is negative if congestion reverses. The data patterns above yield several actionable insights:

- **1. On-peak weekday FTRs capture the largest expected congestion rents.** The standard deviation of LMPs—which measures the magnitude of node price dispersion—peaks sharply during weekday morning and evening peak hours. These are the periods when transmission constraints bind most, creating large spreads that an FTR owner can harvest. Holding an FTR on a congested weekday-peak path is analogous to owning an option on the constraint: the right generator-to-load direction earns the shadow price on the binding line.
- **2. Off-peak and weekend FTRs offer lower expected payoff and lower variance.** During overnight and weekend hours the LMP standard deviation collapses, meaning node-to-node spreads shrink. FTRs covering these periods generate modest expected returns. They may still be valuable as hedges for load-serving entities that buy power on weekends (e.g., data centers operating 24/7), but speculators seeking maximum congestion rent should concentrate nominations on peak weekday hours.
- **3. Negative LMPs create directional risk and potential short-side opportunities.** When minimum LMPs are negative—most likely overnight on weekends—FTR positions can reverse sign. A holder of a long FTR (expecting the load-side node to be more expensive) earns a loss if generation-side prices go negative and load-side prices remain positive. Conversely, a *short* FTR, or an FTR on a path from a wind-rich export node to a load center during off-peak hours, could capture the negative-price spread. Market participants should consider the directionality and seasonality of congestion carefully before bidding in MISO's annual and monthly FTR auctions.
- **4. February is a winter-demand month: implications for seasonal FTRs.** February 2025 includes cold periods that elevate heating-driven electricity demand in MISO's Midwest footprint, particularly during weekday mornings. This amplifies the morning peak and could widen congestion on north–south and east–west power corridors. FTRs acquired for winter peak periods may command a seasonal premium. Holders of annual FTRs must account for this seasonality: the same path could be very valuable in winter peak hours but nearly worthless during summer overnight hours.
- **5. The morning ramp (HE 6–9) deserves particular attention.** The rapid weekday LMP escalation between HE 6 and HE 9 mirrors the Duck Curve ramp discussed in the slides—demand rises steeply while renewable generation is still ramping up. As wind and solar penetration grow in MISO, this ramp steepness is expected to intensify. FTRs covering the morning ramp hours on paths from generation-constrained areas to load centers may increase in value over time, offering an attractive long-term position for investors who anticipate continued renewable build-out.